Wireshark Display Filter Analysis Report

Course: Cyber Security Lab CSE-4174
Experiment: Understanding Wireshark Display Filters
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Objective

The objective of this experiment was to study different Wireshark display filters and understand the network traffic they display. Our group tested the `udp` and `tls or ssl` filters.

Filter 1: `udp`

When we applied the `udp` filter, Wireshark showed only UDP packets and hid the other protocols. UDP is a protocol that sends data without making a connection first, so it is fast but does not guarantee that every packet will reach the destination.

In our capture, we saw different UDP-based protocols like **DNS**, **NBNS**, **LLMNR**, **SSDP**, and **DHCPv6**.

One interesting packet was a DNS request from **172.16.13.39** to **8.8.8.8**, which is Google's DNS server. The computer was trying to find the IP address of **array507.prod.do.dsp.mp.microsoft.com**. After that, the DNS server replied with the correct IP address. We also noticed NBNS and LLMNR packets, which are used by Windows to find devices on the local network.

Filter 2: `tls or ssl`

Next, we used the `tls or ssl` filter. This filter showed only encrypted traffic.

We could see the TLS handshake, including **Client Hello**, **Server Hello**, **Certificate**, and **Key Exchange** packets. After these steps, the packets changed to **Application Data**, which means the connection became encrypted.

communication was between our computer (**172.16.13.39**) and a Microsoft server (**72.153.5.128**) using **TCP port** which is the standard port for HTTPS. We also noticed some **QUIC** packets, which are used by modern web wsers for HTTP/3 connections.

Conclusion

From this experiment, we learned that the `udp` filter is useful for finding protocols like DNS and NBNS, while the `tls or ssl` filter helps us see how secure connections are created. Using these filters makes it much easier to understand network traffic and focus only on the packets we want to analyze.